

# Jose Luis Ponton

GENERATIVE AI · HUMAN MOTION · ROBOTICS · COMPUTER GRAPHICS · XR/VR

Saarbrücken, 66111, Germany

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## Research Experience

### Max Planck Institute for Informatics

Saarbrücken, Germany

POSTDOCTORAL RESEARCHER

May 2026 - present

- Robotics, Generative AI and Human Motion Models.
- Supervisor: Christian Theobalt.

### Universitat Politecnica de Catalunya

Barcelona, Spain

POSTDOCTORAL RESEARCHER

Feb. 2026 - Apr. 2026

- Researched generative AI for virtual humans animation.
- Co-supervised PhD and Master students in computer graphics and contributed to the preparation of competitive European grant proposals.
- Supervisors: Nuria Pelechano and Carlos Andujar.

### Meta

Zurich, Switzerland

RESEARCH SCIENTIST INTERN

Jun. 2025 - Jan. 2026

- Developed a scalable AI motion style transfer framework; from prototype to production within the metaverse team (SIGGRAPH 2026 paper).
- Supervisors: Petr Kadleček and Yuting Ye.

### Max Planck Institute for Informatics

Saarbrücken, Germany

RESEARCH INTERN

May 2024 - Jul. 2024

- Researched diffusion model-based motion capture from insole sensor data (Eurographics 2026 paper).
- Supervisors: Christian Theobalt and Marc Habermann.

### Universitat Politecnica de Catalunya

Barcelona, Spain

FPU PRE-DOCTORAL RESEARCHER (FPU21/01927)

Jan. 2023 - Jun. 2025

- Awarded a highly competitive fellowship from the Spanish Ministry of Science (Ranked 11th of 32 nationwide grants in Computer Science).

### Universitat Politecnica de Catalunya

Barcelona, Spain

RESEARCHER

May 2021 - Dec. 2022

- Researched real-time data-driven motion synthesis for character animation such as Motion Matching.

### CYENS Centre of Excellence

Nicosia, Cyprus

RESEARCH INTERN

Jul. 2022 - Sep. 2022

- Researched human motion reconstruction from VR sparse input using deep learning (SIGGRAPH Asia 2023 paper).
- Supervisor: Andreas Aristidou.

### Universitat Politecnica de Catalunya

Barcelona, Spain

RESEARCH INTERN

Apr. 2019 - Apr. 2021

- Developed a VR-assisted architectural design application for the architects of Sagrada Familia.

## Teaching & Additional Experience

### Universitat Politecnica de Catalunya

Barcelona, Spain

TEACHING ASSISTANT

Sep. 2023 - Jun. 2025

- Taught Computer Graphics and GPU Programming in the Computer Science degree.

### Hyper Online (Y Combinator startup)

(Remote) San Francisco, California, USA

RESEARCH ENGINEER - FREELANCE

Feb. 2024 - Jan. 2025

- Developed the computer vision-based motion capture pipeline for Hyper's desktop application. Integration and post-processing of computer vision-based pose, hand and face tracking techniques, and retargeting to skeletal models.
- Researched AI models for audio to blendshapes / morph targets.

## Education

### Universitat Politècnica de Catalunya

PH.D. IN COMPUTING

- Grade: Cum Laude. Awarded the best PhD award by the Eurographics Spanish Section.
- Title: *Learning Data-driven Character Animation for Avatars in Virtual Reality*
- Supervisors: Nuria Pelechano and Carlos Andujar.

Barcelona, Spain

Dec. 2022 - Dec. 2025

### Universitat Politècnica de Catalunya

M.SC. IN INNOVATION AND RESEARCH IN INFORMATICS

- Grade: 9.15/10.00
- Specialization in Computer Graphics, Character Animation and Virtual Reality

Barcelona, Spain

Sep. 2020 - Jul. 2022

### Universitat Politècnica de Catalunya

B.S. IN COMPUTER SCIENCE AND ENGINEERING

- Grade: 9.03/10.00 (Rank 3 of 239).
- Specialization in Artificial Intelligence, 3D Graphics/Geometry and Game Programming.

Barcelona, Spain

Sep. 2016 - Jul. 2020

## Selected Publications

### STyMo: Fast and Controllable Few-Shot Motion Style Transfer

JOSE LUIS PONTON, ALEXANDER WINKLER, LADISLAV KAVAN, YUTING YE, PETR KADLECEK

SIGGRAPH 2026. July 2026. Los Angeles. USA.

ACM Transactions on Graphics (10.1145/3811356).

### Step2Motion: Locomotion Reconstruction from Pressure Sensing Insoles

JOSE LUIS PONTON, EDUARDO ALVARADO, LIN GENG FOO, NURIA PELECHANO, CARLOS ANDUJAR, MARC HABERMANN

Accepted at Eurographics 2026. May 2026. Aachen. Germany.

Computer Graphics Forum (10.1111/cgf.70405).

### Monkey See, Monkey Break? Study of Rule-Breaking Imitation in Virtual Crowds

KEXIANG HUANG, TAIRAN YIN, JOSE LUIS PONTON, RUIDA TANG, REIYA ITATANI, GANGYI DING, OSCAR ARGUDO, NURIA PELECHANO

IEEE VR 2026 · 33rd IEEE Conference on Virtual Reality and 3D User Interfaces. March 2026. Daegu. South Korea.

IEEE Transactions on Visualization and Computer Graphics (10.1109/TVCG.2026.3679141).

### Environment-aware Motion Matching

JOSE LUIS PONTON, SHELDON ANDREWS, CARLOS ANDUJAR, NURIA PELECHANO

SIGGRAPH Asia. Dec. 2025. Hong Kong. China.

ACM Transactions on Graphics (10.1145/3763334).

### DragPoser: Motion Reconstruction from Variable Sparse Tracking Signals via Latent Space Optimization

JOSE LUIS PONTON, EDUARD PUJOL, ANDREAS ARISTIDOU, CARLOS ANDUJAR, NURIA PELECHANO

Eurographics. May. 2025. London, UK.

Computer Graphics Forum (10.1111/cgf.70026).

### Stretch your reach: Studying Self-Avatar and Controller Misalignment in Virtual Reality Interaction

JOSE LUIS PONTON, REZA KESHAVARZ, ALEJANDRO BEACCO, NURIA PELECHANO

CHI conference on Human Factors in Computing Systems. May. 2024. Hawaii, USA. (10.1145/3613904.3642268)

### Exploring the Role of Expected Collision Feedback in Crowded Virtual Environments

HAORAN YUN, JOSE LUIS PONTON, ALEJANDRO BEACCO, CARLOS ANDUJAR, NURIA PELECHANO

IEEE VR 2024 · 31th IEEE Conference on Virtual Reality and 3D User Interfaces. Mar. 2024. Orlando, USA. (10.1109/VR58804.2024.00068)

### SparsePoser: Real-time Full-body Motion Reconstruction from Sparse Data

JOSE LUIS PONTON, HAORAN YUN, ANDREAS ARISTIDOU, CARLOS ANDUJAR, NURIA PELECHANO

SIGGRAPH Asia. Dec. 2023. Sydney. Australia.

ACM Transactions on Graphics (10.1145/3625264).

### Fitted avatars: automatic skeleton adjustment for self-avatars in virtual reality

JOSE LUIS PONTON, VÍCTOR CEBALLOS, LESLY ACOSTA, ALEJANDRO RÍOS, EVA MONCLÚS, NURIA PELECHANO

Virtual Reality. Springer. Jul. 2023. (10.1007/s10055-023-00821-z)

## Animation Fidelity in Self-Avatars: Impact on User Performance and Sense of Agency

HAORAN YUN, **JOSE LUIS PONTON**, CARLOS ANDUJAR, NURIA PELECHANO

IEEE VR 2023 · 30th IEEE Conference on Virtual Reality and 3D User Interfaces. Mar. 2023. Shanghai, China. (10.1109/VR55154.2023.00044)

## Combining Motion Matching and Orientation Prediction to Animate Avatars for Consumer-Grade VR Devices

**JOSE LUIS PONTON**, HAORAN YUN, CARLOS ANDUJAR, NURIA PELECHANO

SCA 2022 · ACM SIGGRAPH/Eurographics Symposium on Computer Animation. Sep. 2022. Durham, United Kingdom.

Computer Graphics Forum (10.1111/cgf.14628). *Top 10% viewed paper 2023.*

## AvatarGo: Plug and Play self-avatars for VR

**JOSE LUIS PONTON**, EVA MONCLUS, NURIA PELECHANO

Eurographics - Short Papers. May. 2022. Reims, France. (10.2312/egs.20221037).

## Academic Service

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2026 **IPC Member**, 25th ACM SIGGRAPH / Eurographics Symposium on Computer Animation (SCA)

*Barcelona, Spain*

2026 **IPC Member**, XXXV Spanish Conference on Computer Graphics (CEIG)

*Valencia, Spain*

2025 **IPC Member**, 18th ACM SIGGRAPH conference on Motion, Interaction and Games (MIG)

*Zurich, Switzerland*

2024 **IPC Member**, CLIPE Workshop Creating Lively Interactive Populated Environments

*Limassol, Cyprus*

**Reviewer**, ACM SIGGRAPH, ACM SIGGRAPH Asia, ACM CHI, IEEE TVCG, Eurographics, IEEE VR, Computer Graphics Forum, Computer & Graphics, Computer Graphics International, International Journal of Human-Computer Interaction, MIG, ACM SCA, Springer Virtual Reality, CEIG

## Open Source Contributions

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### Motion Matching (over 500 stars on GitHub)

Motion Matching implementation for the Unity game engine. It includes a spring-based character controller, BVH-accelerated search, inertialization blending, automatic and customizable feature extraction/selection, foot lock, tags, among other features.

<https://github.com/JLPM22/MotionMatching>

### PyMotion

Python library for manipulating and **processing motion data** in NumPy or **PyTorch**. It is designed to facilitate the development of neural networks for character animation. Features: quaternion operations and conversions to other rotation representations; dual quaternions; BVH importer/exporter; skeletal operations such as forward kinematics; Blender integration. <https://github.com/UPC-ViRVIG/pymotion>

### Game Development Educator & YouTube Creator

Established and grew a large Spanish-language YouTube channel on game development, achieving **over 120,000 subscribers and 9M+ views**.

It involved creating technical tutorials and simplified explanations on topics like gaming, computer graphics, character animation, and AI, demonstrating communication for a global audience. <https://www.youtube.com/c/JLPMGameDev/videos>